AHMED ALI, MS

Data Analyst | Data Scientist | Machine Learning

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LinkedIn: https://www.linkedin.com/in/ahmed-ali-348236357/ **| Portfolio:** https://ahmedali-ds.github.io/

Work Authorization: F-1 OPT with valid EAD card, authorized to work in the U.S. (STEM OPT eligible)

Professional Summary

Data Science graduate (M.S., May 2026) with hands-on experience building and evaluating forecasting and classification models in Python, spanning data preparation, pattern and anomaly identification, feature engineering, and model evaluation and optimization. Combines applied machine learning skills with three years of production software engineering experience writing scalable code and optimizing systems for performance and reliability. Skilled in turning model results into practical recommendations for technical and non-technical stakeholders, with a strong foundation in documentation and process standardization.

Professional Experience

AppCrates | Pakistan

PHP Laravel Developer | 2020 to 2023

- Wrote and optimized complex SQL queries (multi-table joins, aggregations, subqueries) to power a real-time order-processing system, improving performance and reliability
- Built and maintained backend features and RESTful API endpoints for 3+ production applications, supporting live deployments
- Documented system behavior and query logic while optimizing database indexing and query structure, improving data retrieval efficiency in a live production environment

Skills

- **Programming:** Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), SQL (MySQL), R
- **Machine Learning & Modeling:** Forecasting Models, Classification, Regression, Logistic Regression, Random Forest, Clustering (K-Means, Hierarchical), Feature Engineering, Classification Threshold Tuning, Algorithm Optimization, Baseline Model Comparison, Model Evaluation (Accuracy, F1, RMSE, MAE, MAPE, AUC)
- **Data Preparation:** Data Cleaning, Missing-Value Imputation, Categorical Encoding
- **Statistics & Analysis:** Statistical Inference, Hypothesis Testing, EDA, Pattern & Anomaly Identification
- **Production & Deployment:** Production Code, RESTful APIs, SQL Query Optimization, Performance Optimization, Git/GitHub
- **Reporting & Documentation:** Power BI (basic), Folium, Excel (PivotTables, advanced formulas), Documentation

Key Projects

AustinEasyPark | Parking Availability Prediction System (In Progress) | Python, Random Forest, Folium

- Building an end-to-end ML pipeline predicting block-level street parking availability in downtown Austin using real municipal transaction data from the City of Austin
- Engineering features from time patterns, weather signals, and local event schedules to train a Random Forest classifier that predicts block-level occupancy
- Developing interactive geospatial visualizations using Folium to communicate parking availability patterns across downtown Austin

Sales Revenue Forecasting | Python, Regression, Time-Series Analysis, Matplotlib

- **Objective:** Forecast sales revenue from three years of historical sales data and identify recurring patterns and anomalies.
- **Methodology:** Decomposed trend and seasonality, engineered lag features and rolling statistics, and evaluated models with MAE, RMSE, and MAPE, iterating to improve accuracy.
- **Quantitative Result:** Reduced RMSE by 22% versus a moving-average baseline.
- **GitHub:** github.com/AhmedAli-DS/sales-forecasting

Customer Churn Prediction | Python, Scikit-learn, Pandas, Matplotlib

- **Objective:** Identify at-risk customers within an imbalanced dataset of 7,000+ rows.
- **Methodology:** Built a pipeline with data cleaning, missing-value imputation, and encoding of 15 categorical features; trained Logistic Regression and Random Forest models with threshold tuning; analyzed feature importance to surface key drivers.
- **Quantitative Result:** Achieved 87% accuracy and 0.83 F1-score on the minority class, and turned model output into practical recommendations for non-technical stakeholders.
- **GitHub:** github.com/AhmedAli-DS/Telecom-customer-churn--Model-Evaluation

Education

Master of Science in Data Science | May 2026
Lindsey Wilson University, Columbia, KY | GPA 3.67

Bachelor of Science in Computer Science | 2020
University of Central Punjab, Pakistan